REPORT

Final Capstone Report: Social Network Explorer (SNE)

1. Brief Design Explanation

The Social Network Explorer (SNE) is a command-line application designed to model real-world social interactions

- **User Profiles**: Managed using **Hashing** (via Python dictionaries) to ensure O(1) average time complexity for adding and retrieving user data.
- **Connection Network**: Modeled as a **Graph using an Adjacency List**, allowing for memory-efficient storage of bidirectional friendships.
- Discovery Engines: Uses BFS to find the "Shortest Path" (degrees of separation) and DFS for exploring "Friends-of-Friends" within a specific depth
- Recommendations: A score-based system that implements Insertion Sort to rank potential friends by common interest counts.

2. Complexity Notes

Operation	Data Structure / Algorithm	Time Complexity	Reason
Add/Get/Update User	HashTable (Dictionary)	$O(1)$	Direct key-based access.
BFS Shortest Path	Queue-based Traversal	$O(V+E)$	Visits each user (V and connection (E once.
DFS Depth Search	Recursive Traversal	$O(V+E)$	Explores branches to limit k.
Sorting Suggestions	Insertion Sort	$O(n^2)$	Comparing n candidates against each other.

3. Conceptual Recommendation Discussion

In modern social platforms, recommendation engines rely on the synergy between hashing and sorting

- **Hashing** allows us to group users by shared attributes (like interests) nearly instantaneously. By mapping an interest (e.g., "Robotics") to a list of User IDs, we can filter thousands of candidates into a small relevant subset in O(1) or O(k) time.
- **Common Interest Scoring** is then applied to this subset by calculating the intersection of interest lists between the target user and candidates.
- **Sorting** (such as the implemented Insertion Sort) is the final layer that organizes these candidates in descending order of their score. This ensures that the most relevant "top 5" suggestions are presented to the user first, optimizing user engagement.

Sample CLI Outputs

--- STARTING AUTOMATED DEMO MODE ---

Added 7 users.

Created 8 friendships.

[BFS] Path from Alice (u1) to Frank (u6):

['u1', 'u4', 'u6']

[BFS] Path from Charlie (u3) to Grace (u7):

['u3', 'u5', 'u7']

[DFS] Friends reachable from Alice (u1) at Depth 2:

{'u1', 'u3', 'u5', 'u7', 'u4'}

[Suggestions] Top friend suggestions for Alice (u1):

User: u2, Common Interests: 0

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========================================
   LOADING DEMO DATA...
Demo Data Ready.
========================================

--- SOCIAL NETWORK EXPLORER ---
1: Add | 2: View | 3: Update | 4: Friend | 5: BFS | 6: DFS | 7: Recs | 8: Exit

Select a number: 1
ID: 123
Name: raju
Age: 16
Interests (comma): games
>> Added.

--- SOCIAL NETWORK EXPLORER ---
1: Add | 2: View | 3: Update | 4: Friend | 5: BFS | 6: DFS | 7: Recs | 8: Exit

Select a number: 2
Enter ID to View: 123

FOUND: raju, Age: 16, Interests: ['games']

--- SOCIAL NETWORK EXPLORER ---
1: Add | 2: View | 3: Update | 4: Friend | 5: BFS | 6: DFS | 7: Recs | 8: Exit

Select a number: 3
ID to Update: 123
New Name: rahul
>> Updated.

--- SOCIAL NETWORK EXPLORER ---
1: Add | 2: View | 3: Update | 4: Friend | 5: BFS | 6: DFS | 7: Recs | 8: Exit

Select a number: 4
ID 1: 123
ID 2: 1234
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--- SOCIAL NETWORK EXPLORER ---
1: Add | 2: View | 3: Update | 4: Friend | 5: BFS | 6: DFS | 7: Recs | 8: Exit

Select a number: 5
Start ID: 123
End ID: 1234
>> Path: []

--- SOCIAL NETWORK EXPLORER ---
1: Add | 2: View | 3: Update | 4: Friend | 5: BFS | 6: DFS | 7: Recs | 8: Exit

Select a number: 6
ID: 1
Depth: 4
>> Reachable: set()

--- SOCIAL NETWORK EXPLORER ---
1: Add | 2: View | 3: Update | 4: Friend | 5: BFS | 6: DFS | 7: Recs | 8: Exit

Select a number: 7
User ID: 123
>> Recommendations: []

--- SOCIAL NETWORK EXPLORER ---
1: Add | 2: View | 3: Update | 4: Friend | 5: BFS | 6: DFS | 7: Recs | 8: Exit

Select a number: 8
Goodbye!
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